Malachi Mabie

AI Security and Machine Learning Engineer

Carpinteria, CA | Open to relocation to Los Angeles South Bay and Orange County

malachimabie.com | linkedin.com/in/mlmabie | github.com/mlmabie

Summary

AI/ML engineer with defense research experience in vulnerability repair, behavior analytics, and cognitive security. Builds independent checks for agent behavior and model outputs. Current AMR work applies this approach to robotics software, evaluation, and runtime integration.

Experience

Advanced Metal Research / Gradient Industrial Robotics | July 2026 to Present

Robotics ML Research Engineer (contract)

- Build agent-assisted robotics software spanning motion-planning evaluation, typed task records, candidate trajectories, independent checks, and replayable experiment evidence.
- Develop Steam Deck operator controls and robot runtime integration. Resolved packet, identity, and configured-axis mismatches, contributing to operator-confirmed jogging on the assembled robot.
- Tie evaluation results to code revisions and robot frames, with independent kinematic, collision, calibration, and interface checks. Keep jogging, full trajectory execution, and welding validation separate.

Mabie Industries, LLC | April 2025 to Present

AI Software Engineer / Independent Contractor

- Built agent research infrastructure with typed tools, retrieval, memory, execution traces, replay, and independent verifiers. Delivered cognitive-security threat modeling and evaluation for Siethar Group.
- Built and shipped six production systems for Drone Institute, including thermal imagery and GIS workflows, a marketplace, partner onboarding, operations, and order fulfillment. Owned architecture through deployment and production debugging.

RAM Laboratories, Inc. | April 2022 to March 2025

Senior Member of Technical Staff / Principal Investigator, AI/ML R&D

- Directed VULCAN for the DARPA AI Cyber Challenge, combining code models with static analysis, fuzzing, and runtime verification to evaluate generated patches.
- Invented DL-Patcher for automated C/C++ vulnerability repair, with synthetic bug insertion, training-data filtering, checker feedback, and human review. U.S. patent application 18/375,839 filed.
- Led T-UEBA, a $250,000 Army SBIR for tactical user and entity behavior analytics. Reported program evaluation results of 0.887 F1, under 3% false positives, and 15.2 ms per graph with a 15 MB temporal graph model.

Applied Medical | June 2013 to September 2018

Polymer Design and Mechanical Engineer; previously Process Development Intern

- Designed and qualified robotic end-of-arm tooling, pneumatic mechanisms, and manufacturing fixtures. A production insert-molding process fix reduced operating variance by approximately $100,000 per month.

Education

University of California San Diego | B.S. Cognitive Science, 2021

Specialization in Machine Learning and Neural Computation.

Technical skills

Python, C/C++, PyTorch, TensorFlow, SQL; vulnerability repair, static analysis, fuzzing, runtime verification, adversarial evaluation, threat modeling, graph learning; AWS, Docker, Linux.